

EXTENSION 1

Stage 6

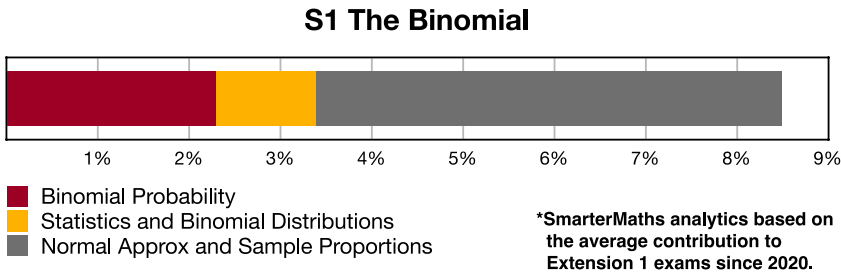
Statistical Analysis (Ext1), S1 The Binomial (Y12)

Normal Approximations of Sample Proportions (Ext1)

Teacher: SHS Maths

Exam Equivalent Time: 96 minutes (based on allocation of 1.5 minutes per mark)





HISTORICAL CONTRIBUTION

- S1 The Binomial* has contributed an average of 8.6% per new syllabus exam since it was introduced in 2020.
- This topic has been split into three categories for analysis purposes which are: *1-Binomial Probability* (2.3%), *2-Statistics and Binomial Distributions* (1.2%) and *3-Normal Approximations of Sample Proportions* (5.1%).
- This analysis looks at *Normal Approximations of Sample Proportions*.

HSC ANALYSIS - What to expect and common pitfalls

- Normal Approximations of Sample Proportions* (5.1%) has appeared in the longer answer section of every new syllabus Ext1 exam to date, including twice in 2022.
- This topic combines significant mark allocations with high difficulty → critical revision area.
- Proficiency in understanding and using a probability table of z-scores is clearly becoming a critical skill (required in 2024 Ext1 12c, 2023 Ext1 11f and 2022 Ext1 14d). Note that all worksheets in this subtopic automatically include this reference table.
- Supplementing the HSC questions on this new topic, the question database is informed by specific syllabus dot points, *NESA Topic Guidance*, official NESA "sample" exams and other States who have an exam history of covering similar content.
- Understand limitations to the normal approximation (see *EXT1 S1 EQ-Bank 21*).

Questions

1. Statistics, EXT1 S1 2023 HSC 2 MC

A standard six-sided die is rolled 12 times.

Let $\hat{p}$ be the proportion of the rolls with an outcome of 2.

Which of the following expressions is the probability that at least 9 of the rolls have an outcome of 2?

- A.

$P\left(\hat{p} \geq \frac{3}{4}\right)$
- B.

$P\left(\hat{p} \geq \frac{1}{6}\right)$
- C.

$P\left(\hat{p} \leq \frac{3}{4}\right)$
- D.

$P\left(\hat{p} \leq \frac{1}{6}\right)$

2. Statistics, EXT1 S1 2024 HSC 8 MC

A local council is proposing to ban dog-walking on the beach. It is known that the proportion of households that have a dog is $\frac{7}{12}$.

The local council wishes to poll n households about this proposal.

Let $\hat{p}$ be the random variable representing the proportion of households polled that have a dog.

What is the smallest sample size, n , for which the standard deviation of $\hat{p}$ is less than 0.06?

- A.

67
- B.

68
- C.

94
- D.

95

3. Statistics, EXT1 S1 2020 HSC 12b

When a particular biased coin is tossed, the probability of obtaining a head is $\frac{3}{5}$.

This coin is tossed 100 times.

Let X be the random variable representing the number of heads obtained. This random variable will have a binomial distribution.

- i.

Find the expected value, $E(X)$. (1 mark)
- ii.

By finding the variance, $\text{Var}(X)$, show that the standard deviation of X is approximately 5. (1 mark)
- iii.

By using a normal approximation, find the approximate probability that X is between 55 and 65. (1 mark)

4. Statistics, EXT1 S1 EQ-Bank 20

- Netball Australia records show that 10% of all registered players are over the age of 25.
- i. A random survey of 100 netball players was carried out to find out how many were over 25 years of age. Assuming the sample proportion is normally distributed, calculate the expected mean and standard deviation of this group. *(2 marks)*
- ii. Using the probability table attached, estimate the probability that at least 15 players surveyed will be over 25 years of age. *(2 marks)*

5. Statistics, EXT1 S1 2023 HSC 11f

- A recent census found that 30% of Australians were born overseas.
- A sample of 900 randomly selected Australians was surveyed.
- Let $\hat{p}$ be the sample proportion of surveyed people who were born overseas.
- A normal distribution is to be used to approximate $P(\hat{p} \leq 0.31)$.
- i. Show that the variance of the random variable $\hat{p}$ is $\frac{7}{30\,000}$. *(2 marks)*
- ii. Use the standard normal distribution and the normal distribution table to approximate $P(\hat{p} \leq 0.31)$, giving your answer correct to two decimal places. *(2 marks)*

6. Statistics, EXT1 S1 2024 HSC 12c

- A charity employs a worker to collect donations. There is a 0.31 chance that when the charity worker talks to someone a donation is made to the charity.
- Each day the charity worker must talk to exactly 100 people.
- Use a standard normal distribution table to approximate the probability that, on a particular day, at least 35% of the people talked to made a donation. *(3 marks)*

7. Statistics, EXT1 S1 SM-Bank 16

- For a certain species of bird, the proportion of birds with a crest is known to be $\frac{3}{5}$.
- Let $\hat{p}$ be the random variable representing the proportion of birds with a crest in samples of size n for this specific bird.
- Find the smallest sample size for which the standard deviation of $\hat{p}$ is less than 0.08. *(2 marks)*

8. Statistics, EXT1 S1 EQ-Bank 22

- It is known that 43% of voters in a city voted for Labor in the election.
- If 250 voters from the city are selected at random, use a suitable approximation and the attached probability table to find the probability that at least 120 of those chosen voted for Labor. *(3 marks)*

9. Statistics, EXT1 S1 EQ-Bank 24

- The proportion of a population that have brown eyes is 0.35.
- Using $\hat{p}$ to represent the sample proportion, calculate the smallest sample size, n , such that the standard deviation of $\hat{p}$ is below 0.02. *(2 marks)*

10. Statistics, EXT1 S1 SM-Bank 5

- A manufacturer makes torches that have a probability of 0.03 of being defective.
- Let $\hat{p}$ be the random variable that represents the sample proportion of torches for samples of size n drawn from production.
- Find the smallest integer value of n such that the standard deviation of $\hat{p}$ is less than $\frac{1}{50}$. *(2 marks)*

11. Statistics, EXT1 S1 SM-Bank 6

- After counting votes in an election, it is known that 40% of people voted for the Katter party.
- A sample of 600 voting cards are taken and inspected. Assuming this sample proportion is approximately normally distributed, what is the probability that the percentage of voting cards inspected that chose the Katter party is less than 36%? *(3 marks)*

12. Statistics, EXT1 S1 2017 MET1 4

- In a large population of fish, the proportion of angel fish is $\frac{1}{4}$.
- Let $\hat{p}$ be the random variable that represents the sample proportion of angel fish for samples of size n drawn from the population.
- Find the smallest integer value of n such that the standard deviation of $\hat{p}$ is less than or equal to $\frac{1}{100}$. *(2 marks)*

13. Statistics, EXT1 S1 EQ-Bank 23

- A light manufacturer knows that 6% of the light bulbs it produces are defective.
- Light bulbs are supplied in boxes of 20 bulbs. Boxes are supplied in pallets of 120 boxes.
- Calculate the probability that
- i. A box of light bulbs contains exactly 3 defective bulbs. *(1 mark)*
- ii. A box of light bulbs contains at least 1 defective bulb. *(1 mark)*
- iii. A pallet contains between 90 and 95 (inclusive) boxes with at least 1 defective bulb (use the probability table attached). *(3 marks)*

14. Statistics, EXT1 S1 2019 MET1 6

Jacinta tosses a coin five times.

- a. Assuming that the coin is fair and given that Jacinta observes a head on the first two tosses, find the probability that she observes a total of either four or five heads. (2 marks)
- b. Albin suspects that a coin is not actually a fair coin and he tosses it 18 times.
Albin observes a total of 12 heads from the 18 tosses.

Let X = probability of obtaining a head.

Find the range of X in which 95% of observations are expected to lie within. (2 marks)

15. Statistics, EXT1 S1 2022 HSC 13e

- A chocolate factory sells 150-gram chocolate bars. There has been a complaint that the bars actually weigh less than 150 grams, so a team of inspectors was sent to the factory to check. They randomly selected 16 bars, weighed them and noted that 8 bars weighed less than 150 grams.
- The factory manager claims 80% of the chocolate bars produced by the factory weigh 150 grams or more.
- i. The inspectors used the normal approximation to the binomial distribution to calculate the probability, $\mathcal{P}$, of having at least 8 bars weighing less than 150 grams in a random sample of 16, assuming the factory manager's claim is correct.

Using the attached probability table, calculate the value of $\mathcal{P}$. (2 marks)
 - ii. The factory manager disagrees with the method used by the inspectors as described in part (i).

Explain why the method used by the inspectors might not be valid. (1 mark)

16. Statistics, EXT1 S1 SM-Bank 11

- Within a particular population, it is known that the percentage of left-handers is 17%.
- A research project randomly selects 200 people from this population.
- Assuming this sample proportion is normally distributed, what is the probability that the percentage of people that are left-handed in this sample is
- i. greater than 20% (2 marks)
 - ii. less than 10% (2 marks)

17. Statistics, EXT1 S1 EQ-Bank 21

- A biased coin has a 0.6 chance of landing on heads. The coin is tossed 15 times.
- i. Calculate the probability of obtaining 7, 8 or 9 heads using binomial probability distribution.

Give your answer correct to 3 decimal places. (2 marks)
 - ii. Calculate the probability of obtaining 7, 8 or 9 heads using normal approximation to the binomial distribution and the probability table attached.

Give your answer correct to 3 decimal places. (2 marks)
 - iii. Could this binomial distribution be reasonably approximated with a normal distribution? Support your answer with a brief calculation. (1 mark)

18. Statistics, EXT1 S1 2021 HSC 14d

At a certain factory, the proportion of faulty items produced by a machine is $p = \frac{3}{500}$, which is considered to be acceptable. To confirm that the machine is working to this standard, a sample of size n is taken and the sample proportion $\hat{p}$ is calculated.

It is assumed that $\hat{p}$ is approximately normally distributed with $\mu = p$ and $\sigma^2 = \frac{p(1 - p)}{n}$.

Production by this machine will be shut down if $\hat{p} \geq \frac{4}{500}$.

The sample size is to be chosen so that the chance of shutting down the machine unnecessarily is less than 2.5%.

Find the approximate sample size required, giving your answer to the nearest thousand. (3 marks)

19. Statistics, EXT1 S1 EQ-Bank 14

- It is known that 65% of adults over the age of 60 have been tested for bowel cancer.
- A random sample of 140 adults aged over 60 years is surveyed.
- Using a normal approximation to the binomial distribution and the probability table attached, calculate the probability that at least 85 of the adults chosen have been tested for bowel cancer. (3 marks)

20. Statistics, EXT1 S1 SM-Bank 8

- A laptop's battery is considered faulty if its battery life is less than 3 hours.
- The laptop supplier knows that the chance of a faulty battery in any laptop is 6.5%.
- A random sample of 70 laptops is selected from the supplier and the battery life of each laptop is tested.
- Assuming the sample proportion is normally distributed, what is the probability that the percentage of laptops with faulty batteries lies between 5% and 10%?
- Give your answer to the nearest percentage. (3 marks)

21. Statistics, EXT1 S1 2022 HSC 14d

- An airline company that has empty seats on a flight is not maximising its profit.
- An airline company has found that there is a probability of 5% that a passenger books a flight but misses it. The management of the airline company decides to allow for overbooking, which means selling more tickets than the number of seats available on each flight.
- To protect their reputation, management makes the decision that no more than 1% of their flights should have more passengers showing up for the flight than available seats.
- Given management's decision and using the *attached normal distribution probability table* to find a suitable approximation, find the maximum number of tickets that can be sold for a flight which has 350 seats. (4 marks)

Probability Table for Standard Normal Distribution
Negative z-scores

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

Probability Table for Standard Normal Distribution
Positive z-scores

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

Worked Solutions

1. Statistics, EXT1 S1 2023 HSC 2 MC

Let X = number of rolls ≥ 2

$$X \sim \text{Bin}\left(12, \frac{5}{6}\right)$$

$$\text{If } X = 9 \Rightarrow \hat{p} = \frac{9}{12} = \frac{3}{4}$$

$$\therefore P(X \geq 9) = P\left(\hat{p} \geq \frac{3}{4}\right)$$

$\Rightarrow A$

2. Statistics, EXT1 S1 2024 HSC 8 MC

$$E(\hat{p}) = p = \frac{7}{12}$$

$$\sigma^2 = \frac{p(1-p)}{n}$$

Find n such that $\sigma < 0.06$:

$$\left(\frac{6}{100}\right)^2 > \frac{\frac{7}{12} \times \frac{5}{12}}{n}$$

$$n > \frac{35}{144} \times \frac{100^2}{6^2}$$
$$> 67.5$$

$\Rightarrow B$

Worked Solutions

3. Statistics, EXT1 S1 2020 HSC 12b

i. X = number of heads

$$X \sim \text{Bin}(n, p) \sim \text{Bin}\left(100, \frac{3}{5}\right)$$

$$E(X) = np$$

$$= 100 \times \frac{3}{5}$$

$$= 60$$

ii. $\text{Var}(X) = np(1-p)$

$$= 60 \times \frac{2}{5}$$

$$= 24$$

$$\sigma(x) = \sqrt{24}$$

$$\approx 5$$

iii. $P(55 \leq x \leq 65) \approx P(-1 \leq z \leq 1)$

$$\approx 68\%$$

4. Statistics, EXT1 S1 EQ-Bank 20

i. $E(\hat{p}) = p = 0.1$

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n}$$

$$= \frac{0.1(0.9)}{100}$$

$$= 0.0009$$

$$\sigma(\hat{p}) = \sqrt{0.0009}$$

$$= 0.03$$

ii. $P(\text{at least 15 players are over 25}) = P(\hat{p} \geq 0.15)$

$$\hat{p} \sim N(\mu, \sigma) \sim N(0.1, 0.03)$$

$$P(\hat{p} \geq 0.15) = P\left(z \geq \frac{0.15-0.10}{0.03}\right)$$

$$= P(z \geq 1.67)$$

$$= 1-0.9525$$

$$= 0.0475$$

5. Statistics, EXT1 S1 2023 HSC 11f

i. Let X = number of people born overseas
 $X \sim \text{Bin}(900, 0.3)$
 $\hat{p} = \frac{X}{900}$
 $\text{Var}(X) = np(1 - p) = 900 \times 0.3 \times 0.7$
 $\text{Var}(\hat{p}) = \frac{\text{Var}(X)}{n^2}$
 $\qquad = \frac{900 \times 0.3 \times 0.7}{900^2}$
 $\qquad = \frac{7}{30\,000}$

ii. $\mu = E(\hat{p}) = 0.3, \sigma(\hat{p}) = \sqrt{\frac{7}{30\,000}}$
 $z\text{-score}(0.31) = \frac{0.31 - 0.3}{\sqrt{\frac{7}{30\,000}}} = 0.6546\dots$
 $P(\hat{p} \leq 0.31) = P(z \leq 0.6546\dots)$
 $\qquad = 0.74 \text{ (to 2 d.p.)}$

6. Statistics, EXT1 S1 2024 HSC 12c

$$E(\hat{p}) = p = 0.31, n = 100$$
$$\sigma^2 = \frac{p(1 - p)}{n} = \frac{0.31 \times 0.69}{100} = 0.002139$$
$$\hat{p} \sim N(\mu, \sigma^2) \sim N(0.31, 0.002139)$$
$$P(\hat{p} \geq 0.35) = P\left(Z \geq \frac{x - \mu}{\sigma}\right)$$
$$= P\left(Z \geq \frac{0.35 - 0.31}{\sqrt{0.002139}}\right)$$
$$= P(Z \geq 0.87)$$
$$= 1 - P(Z < 0.87)$$
$$= 1 - 0.8078 \text{ (from table)}$$
$$= 0.1922$$

7. Statistics, EXT1 S1 SM-Bank 16

$$E(\hat{p}) = p = 0.6$$
$$\text{Var}(\hat{p}) = \frac{0.6(1-0.6)}{n} = \frac{0.24}{n}$$
$$\sigma(\hat{p}) = \sqrt{\frac{0.24}{n}}$$

If $\sigma(\hat{p}) < 0.08$,

$$\sqrt{\frac{0.24}{n}} < 0.08$$
$$0.0064n > 0.024$$
$$n > \frac{0.24}{0.0064}$$
$$n > 37.5$$

$\therefore$ Smallest $n = 38$

8. Statistics, EXT1 S1 EQ-Bank 22

$$E(\hat{p}) = p = 0.43$$
$$\text{Var}(\hat{p}) = \frac{0.43(1-0.43)}{250} = 0.0009804$$
$$\sigma(\hat{p}) = 0.0313$$

Let X = number who voted Labor

If $X = 120 \Rightarrow \hat{p} = \frac{120}{250} = 0.48$

$$P(X \geq 120) = P\left(z \geq \frac{0.48-0.43}{0.0313}\right)$$
$$= P(z \geq 1.58)$$
$$= 1-0.9429$$
$$= 0.0571$$

9. Statistics, EXT1 S1 EQ-Bank 24

$$E(\hat{p}) = p = 0.35$$

$$\text{Var}(\hat{p}) = \frac{0.35(1-0.35)}{n} = \frac{0.2275}{n}$$

$$\sigma(\hat{p}) = \sqrt{\frac{0.2275}{n}}$$

$$\text{If } \sigma(\hat{p}) < 0.02,$$

$$\sqrt{\frac{0.2275}{n}} < 0.02$$

$$0.0004n > 0.0275$$

$$n > \frac{0.2275}{0.0004}$$

$$n > 568.75$$

$$\therefore \text{Smallest } n = 569$$

10. Statistics, EXT1 S1 SM-Bank 5

Let X = number of defective torches

$$X \sim \text{Bin}(n, p) \sim \text{Bin}(n, 0.03)$$

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n}$$

$$\sigma(\hat{p}) = \sqrt{\frac{0.03(0.97)}{n}}$$

$$= \sqrt{\frac{291}{10\,000n}}$$

$$\sqrt{\frac{291}{10\,000n}} < \frac{1}{50}$$

$$\frac{291}{10\,000n} < \frac{1}{2500}$$

$$10\,000n > 2500 \times 291$$

$$n > \frac{2500 \times 291}{10\,000}$$

$$> 72.75$$

$$\therefore n_{\min} = 73$$

11. Statistics, EXT1 S1 SM-Bank 6

$$\mu = E(\hat{p}) = p = 0.4$$

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n}$$

$$\sigma^2(\hat{p}) = \frac{0.4(1-0.4)}{600}$$

$$= 0.0004$$

$$\sigma(\hat{p}) = 0.02$$

$$z\text{-score (36\%)} = \frac{0.36 - 0.40}{0.02}$$

$$= -2$$

$$\therefore P(\hat{p} < 36\%) = P(z < -2)$$

$$= 2.5\%$$

12. Statistics, EXT1 S1 2017 MET1 4

Let X = number of angel fish

$$X \sim \text{Bin}\left(n, \frac{1}{4}\right)$$

$$\sigma(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

$$= \sqrt{\frac{\frac{1}{4} \times \frac{3}{4}}{n}}$$

$$= \sqrt{\frac{3}{16n}}$$

$$\text{Solve: } \sqrt{\frac{3}{16n}} \leq \frac{1}{100}$$

$$\frac{3}{16n} \leq \frac{1}{10\,000}$$

$$\frac{30\,000}{16} \leq n$$

$$\therefore n \geq 1875$$

$$\therefore n_{\min} = 1875$$

13. Statistics, EXT1 S1 EQ-Bank 23

- i. $P(D) = 0.06, P(\overline{D}) = 0.94$
- $$P(D = 3) = {}^{20}C_3(0.06)^3(0.94)^{17}$$
- $$= 0.086 \quad (\text{to 3 d.p.})$$
- ii. $P(D \geq 1) = 1 - P(D = 0)$
- $$= 1 - {}^{20}C_0(0.06)^0(0.94)^{20}$$
- $$= 0.710 \quad (\text{to 3 d.p.})$$
- iii. Let X = number of boxes where $D \geq 1$
- Let $\hat{p}$ = proportion of boxes where $D \geq 1$
- $$E(\hat{p}) = p = 0.710$$
- $$\text{Var}(\hat{p}) = \frac{0.710(1-0.710)}{120} = 0.0017158$$
- $$\sigma(\hat{p}) = 0.04142$$
- $$\hat{p} \sim N(0.710, 0.04142)$$
- If $X = 90 \Rightarrow \hat{p} = \frac{90}{120} = 0.75$
- If $X = 95 \Rightarrow \hat{p} = \frac{95}{120} = 0.79167$
- $$z\text{-score } (X = 90) = \frac{0.75 - 0.710}{0.04142} = 0.9657$$
- $$z\text{-score } (X = 95) = \frac{0.79167 - 0.710}{0.04142} = 1.972$$
- $$P(90 \leq X \leq 95) = P(0.97 \leq z \leq 1.97)$$
- $$= 0.9756 - 0.8340$$
- $$= 0.1416$$

14. Statistics, EXT1 S1 2019 MET1 6

- a. After 2 tosses, 2 heads.
- $$P(4 \text{ or } 5 \text{ H}) = HHT + HTH + THH + HHH$$
- $$= \left(\frac{1}{2}\right)^3 \times 4$$
- $$= \frac{1}{2}$$
- b. $E(\hat{p}) = p = \frac{12}{18} = \frac{2}{3}$
- $$\text{Var}(\hat{p}) = \frac{\frac{2}{3}(1 - \frac{2}{3})}{18} = \frac{1}{81}$$
- $$\sigma(\hat{p}) = \frac{1}{9}$$
- 95% interval ($z = \pm 2$)
- $$= (\hat{p} - z \times \sigma(\hat{p}), \hat{p} + z \times \sigma(\hat{p}))$$
- $$= \left(\frac{2}{3} - 2 \times \frac{1}{9}, \frac{2}{3} + 2 \times \frac{1}{9}\right)$$
- $$= \left(\frac{4}{9}, \frac{8}{9}\right)$$

15. Statistics, EXT1 S1 2022 HSC 13e

- i. Let X = number of chocolate bars < 150 grams
- $$X \sim \text{Bin}(16, 0.2)$$
- $$\mu = np = 16 \times 0.2 = 3.2$$
- $$\sigma^2 = np(1 - p) = 16 \times 0.2 \times 0.8 = 2.56$$
- $$z\text{-score } (X = 8) = \frac{8 - 3.2}{\sqrt{2.56}} = 3$$
- $$\mathcal{P}(X \geq 8) = P(z \geq 3.0)$$
- $$= 1 - P(z \leq 3.0)$$
- $$= 1 - 0.9987$$
- $$= 0.0013$$
- ii. $np = 3.2, n(1 - p) = 16 \times 0.8 = 12.8$
- Since $np < 5$, the sample size is too small.

◆ Mean mark part (i) 37%.

Mean mark part (ii) 54%.

16. Statistics, EXT1 S1 SM-Bank 11

i. $E(\widehat{p}) = p = 0.17$

$$\text{Var}(\widehat{p}) = \frac{p(1-p)}{n}$$

$$\sigma^2(\widehat{p}) = \frac{0.17(1-0.17)}{200}$$
$$= 0.0007055\dots$$

$$\sigma(\widehat{p}) = 0.02656\dots$$

$$\widehat{p} \sim N(\mu, \sigma) \sim N(0.17, 0.02656)$$

$$z\text{-score (20\%)} = \frac{x-\mu}{\sigma}$$
$$\approx \frac{0.20-0.17}{0.02656}$$
$$\approx 1.13$$

Using the probability table (attached):

$$\therefore P(\widehat{p} > 20\%) = P(z > 1.13)$$
$$= 1 - 0.8708$$
$$= 0.1292$$

ii. $z\text{-score (10\%)} \approx \frac{0.10 - 0.17}{0.02656}$

$$\approx -2.64$$

Using the probability table (attached):

$$\therefore P(\widehat{p} < 10\%) = P(z < -2.64)$$
$$= 0.0041$$

17. Statistics, EXT1 S1 EQ-Bank 21

i. $P(7, 8 \text{ or } 9) = {}^{15}C_7(0.6)^7(0.4)^8 + {}^{15}C_8(0.6)^8(0.4)^7 + {}^{15}C_9(0.6)^9(0.4)^6$

$$= 0.118056 + 0.177084 + 0.206600$$
$$= 0.502 \quad (\text{to 3 d.p.})$$

ii. $E(\widehat{p}) = p = 0.6$

$$\text{Var}(\widehat{p}) = \frac{p(1-p)}{n} = \frac{0.6 \times 0.4}{15} = 0.016$$

$$\sigma(\widehat{p}) = 0.12649$$

Let X = number of heads

$$P(7, 8, 9) = P(6.5 < X < 9.5)$$

$$\text{If } X = 6.5 \Rightarrow \widehat{p} = \frac{6.5}{15} = 0.4333$$

$$\text{If } X = 9.5 \Rightarrow \widehat{p} = \frac{9.5}{15} = 0.6333$$

$$z\text{-score } (X = 6.5) = \frac{0.4333-0.6}{0.12649} = -1.32$$

$$z\text{-score } (X = 9.5) = \frac{0.6333-0.6}{0.12649} = 0.26$$

$$P(7, 8, 9) = P(-1.32 < z < 0.26)$$
$$= P(z < 1.32) - P(z < -0.26)$$
$$= 0.9066 - 0.3974$$
$$= 0.509$$

iii. $np = 15 \times 0.6 = 9$

$$nq = n(1-p) = 15 \times 0.4 = 6$$

Since $np > 5$ and $nq > 5$,

$\Rightarrow$ Normal approximation is reasonable.

COMMENT: A quick sketch of the normal distribution curve is often helpful when using probability tables in this context.

COMMENT: $np > 10$, $nq > 10$ is also common in assessing if normal approximation is reasonable.

18. Statistics, EXT1 S1 2021 HSC 14d

$$E(\hat{p}) = p = \frac{3}{500} = 0.006$$

$$P\left(\hat{p} \geq \frac{4}{500}\right) < 2.5\%$$

$$\Rightarrow P\left(\hat{p} \geq \frac{4}{500}\right) = P(z\text{-score} \leq 2)$$

$$2 \times \sigma = \frac{4}{500} - E(\hat{p})$$

$$= \frac{4}{500} - \frac{3}{500}$$

$$= \frac{1}{500}$$

$$\sigma = \frac{1}{1000}$$

Using $\sigma^2 = \frac{p(1-p)}{n}$

$$0.001^2 = \frac{0.006(0.994)}{n}$$

$$n = \frac{0.006(0.994)}{0.001^2}$$

$$= 5964$$

$$\approx 6000$$

♦♦ Mean mark 27%.

19. Statistics, EXT1 S1 EQ-Bank 14

Let X = number tested for cancer

$$X \sim \text{Bin}(140, 0.65)$$

Find $P(X \geq 85)$:

$$E(\hat{p}) = p = 0.65$$

$$\text{Var}(\hat{p}) = \frac{0.65(0.35)}{140} = 0.001625$$

$$\sigma(\hat{p}) = 0.04031\dots$$

$$\hat{p} \sim N(\mu, \sigma) \sim N(0.65, 0.04031\dots)$$

If $X = 85 \Rightarrow \hat{p} = \frac{85}{140} = 0.60714\dots$

$$z\text{-score} (X = 85) = \frac{0.60714 - 0.65}{0.04031} \\ \approx -1.063$$

Using the probability table (attached):

$$P(z < -1.06) = 0.1446$$

$$\therefore P(X \geq 85) = 1 - 0.1446 \\ = 0.8554$$

20. Statistics, EXT1 S1 SM-Bank 8

$$E(\hat{p}) = p = 0.065$$

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n}$$

$$\sigma^2(\hat{p}) = \frac{0.065(1-0.065)}{70}$$
$$= 0.0008682\dots$$

$$\sigma(\hat{p}) = 0.029465\dots$$

$$\hat{p} \sim N(\mu, \sigma) \sim N(0.065, 0.029465)$$

$$\text{z-score (5\%)} \approx \frac{0.05-0.065}{0.029465}$$
$$\approx -0.509$$

$$\text{z-score (10\%)} \approx \frac{0.10-0.065}{0.029465}$$
$$\approx 1.187$$

Using probability tables (attached):

$$\therefore P(5\% \leq \hat{p} \leq 10\%) \approx P(-0.51 \leq z \leq 1.19)$$
$$\approx 0.8830 - 0.3050$$
$$\approx 0.5780$$
$$\approx 58\%$$

21. Statistics, EXT1 S1 2022 HSC 14d

Let X = number of passengers taking a flight

$$X \sim \text{Bin}(n, 0.95)$$

$$E(X) = 0.05, \quad \text{Var}(X) = n(0.95)(1-0.95) = 0.0475n$$

$$X \text{ can be approximated by } Y \sim N(0.95n, 0.0475n)$$

Find n such that $P(Y > 350) = 0.01$:

Using the probability table

$\Rightarrow$ z-score of 2.33 corresponds to (closest) upper tail probability < 0.01

$$2.33 = \frac{350 - 0.95n}{\sqrt{0.0475n}}$$

$$2.33\sqrt{0.0475}\sqrt{n} = 350 - 0.95n$$

$$0.95n + 2.33\sqrt{0.0475}\sqrt{n} - 350 = 0$$

$$\sqrt{n} = \frac{-2.33\sqrt{0.0475} \pm \sqrt{\left(2.33\sqrt{0.0475}\right)^2 - 4(0.95)(-350)}}{2(0.95)}$$
$$= 18.9288\dots \quad (n > 0)$$
$$= (18.9288\dots)^2$$
$$\approx 358.30$$

$\therefore$ Maximum tickets that can be sold = 358

◆◆◆ Mean mark 22%.